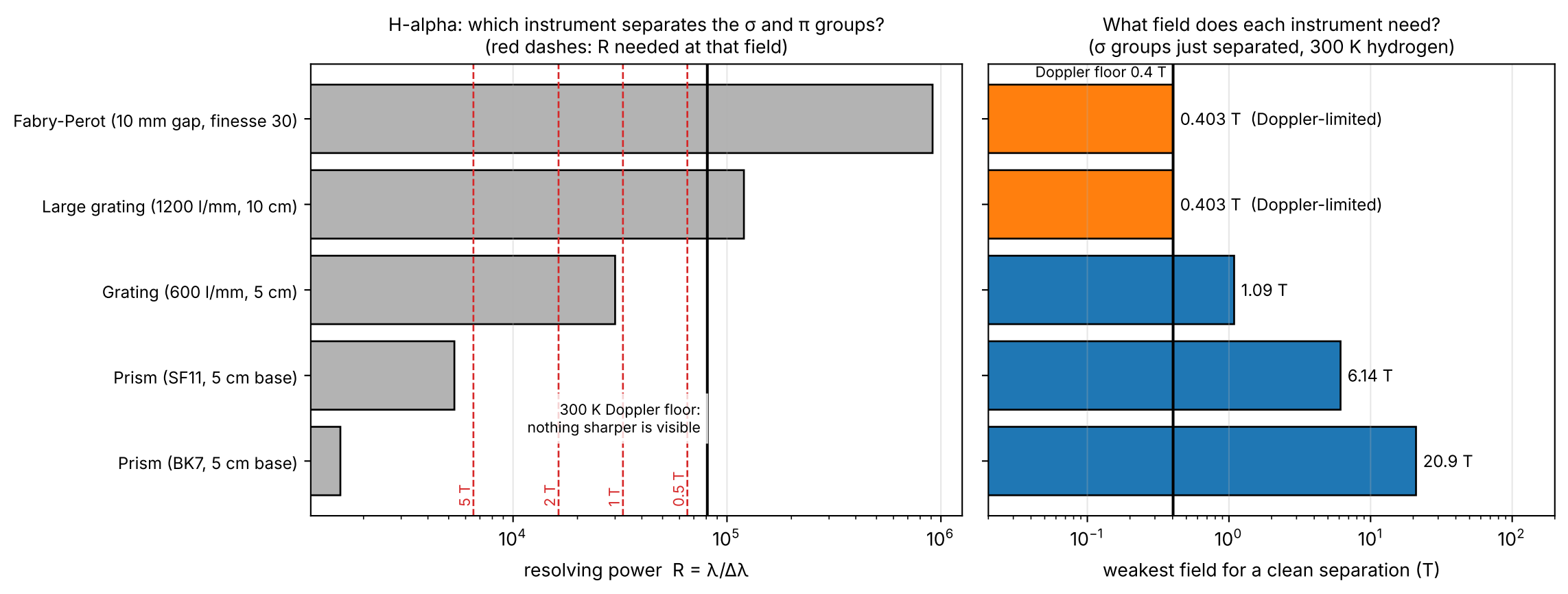
H-alpha: which instrument separates the σ and π groups?
(red dashes: R needed at that field)
What field does each instrument need?
(σ groups just separated, 300 K hydrogen)
Fabry-Perot (10 mm gap, finesse 30)
Large grating (1200 l/mm, 10 cm)
Grating (600 l/mm, 5 cm)
Prism (SF11, 5 cm base)
Prism (BK7, 5 cm base)
300 K Doppler floor:
nothing sharper is visible
5 T
2 T
1 T
0.5 T
Doppler floor 0.4 T
0.403 T (Doppler-limited)
0.403 T (Doppler-limited)
1.09 T
6.14 T
20.9 T
10^4
10^5
10^6
10^{-1}
10^0
10^1
10^2
resolving power R = λ/Δλ
weakest field for a clean separation (T)